Of course, we don’t yet know if such things are possible. While computers are becoming more intelligent each year, so far they have not become more conscious. Intelligence and consciousness are different things. Intelligence is the ability to solve problems, whereas consciousness is the ability to feel things like pain, pleasure, love and hate. In humans and other mammals, intelligence goes hand in hand with consciousness. We solve problems by relying on our feelings. But computers work in a very different way. They may become much more intelligent than us, without ever developing feelings.

On the other hand, we cannot just dismiss the possibility that computers will someday develop consciousness. Why assume that consciousness will always be restricted to organic beings? Is there something magical about carbon-based entities that will forever remain beyond the reach of silicon-based entities? We might soon have the answers to such questions. After 4 billion years of milling around inside the small world of organic compounds, life might suddenly break out into the vastness of the inorganic realm, ready to take up shapes beyond our wildest dreams.

The Singularity

Presently, only a tiny fraction of these new opportunities have been realised. Yet the world of 2014 is already a world in which culture is releasing itself from the shackles of biology. Our ability to engineer not merely the world around us, but above all the world inside our bodies and minds, is developing at breakneck speed. More and more spheres of activity are being shaken out of their complacent ways. Lawyers need to rethink issues of privacy and identity; governments are faced with rethinking matters of health care and equality; sports associations and educational institutions need to redefine fair play and achievement; pension funds and labour markets should readjust to a world in which sixty might be the new thirty. They must all deal with the conundrums of bioengineering, cyborgs and inorganic life.

Mapping the first human genome required fifteen years and $3 billion. Today you can map a person’s DNA within a few weeks and at the cost of a few hundred dollars.[20] The era of personalised medicine – medicine that